

ASHUTOSH ROY

Data Scientist

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SUMMARY

Data Scientist with applied experience in statistical modeling, NLP, and ML pipeline development. Published researcher (RECCAP 2026, IIT Palakkad) with a track record of translating complex datasets into measurable outcomes — from benchmarking OCR systems at IIT Delhi to building predictive models for healthcare data at MitoVoid AI. Skilled in Python, Scikit-learn, Pandas, and Power BI, with strong foundations in feature engineering, evaluation, and visualization.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Conducted **benchmarking and evaluation** of 3 OCR systems (Tesseract, Marker, Gemini OCR) across document types — analyzed failure modes, produced structured comparison metrics, and recommended data-driven extraction strategies.
- Designed a hybrid retrieval pipeline combining semantic vector search (FAISS/Pinecone) and keyword retrieval (Apache Solr); evaluated retrieval precision across thousands of archaeological records.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi. arxiv.org/abs/2511.05667

AI/ML Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Built predictive NLP models for healthcare data analysis; applied **feature engineering** techniques including domain-specific tokenization and clinical entity extraction to improve model performance.
- Developed end-to-end data pipelines: ingestion → preprocessing → feature extraction → model inference, reducing manual processing time significantly.
- Improved patient query resolution time by ~**40%** through optimized intent classification and response generation.

Software Engineer Intern

Nov 2024 – Dec 2024

LeanTactics Solutions Pvt. Ltd.

Remote

- Integrated ML model outputs into production backend systems using Python and FastAPI; optimized data flow between inference layer and application, achieving **25–35% latency reduction**.
- Profiled API performance bottlenecks using data-driven approaches; delivered measurable improvements in request handling at scale.

PROJECTS

Speech Emotion & Stress Detection | Python, PyTorch, Wav2Vec2, BiLSTM, Hugging Face | RECCAP 2026

Jan 2026 – May 2026

- Built multi-task ML model across 3 corpora (RAVDESS, TESS, SAVEE); implemented unified preprocessing (VAD, silence removal, log-Mel spectrograms) and cross-corpus label harmonization for consistent evaluation.
- Achieved **85.9% accuracy, 0.856 F1, RMSE 0.1268** — evaluated using macro-F1, confusion matrix analysis, and regression RMSE; outperformed single-task baselines on all metrics. Paper accepted at RECCAP 2026, IIT Palakkad.

Exam-Helper RAG System | Python, LLaMA 3, FAISS, Pinecone, LangChain

Dec 2024 – Apr 2025

- Designed a retrieval-augmented QA pipeline over large academic corpora; implemented vector indexing and similarity search for context-precise LLM responses with measurably reduced hallucination.

Music Identification App | Python, ACRCLOUD API, Streamlit, NumPy

Sep 2024 – Dec 2024

- Built real-time audio recognition pipeline: audio capture → feature extraction → ACRCLOUD API fingerprinting → metadata retrieval; served via Streamlit UI for instant song identification.

BulkyMail | Python, Streamlit, SMTP, Jinja2, Pandas

Dec 2024

- Built a bulk email automation tool with dynamic Jinja2 templating over recipient data (name, role, company); processed structured CSV lists to drive personalized large-scale outreach campaigns.

Medical Chatbot | Mistral 7B, QLoRA, Tesseract OCR, Transformers

Mar 2024 – Apr 2024

- Fine-tuned Mistral 7B on domain-specific medical datasets; built OCR preprocessing pipeline to extract structured clinical data from unstructured reports for downstream model inference.

TECHNICAL SKILLS

Data Science & ML: Statistical Modeling, Feature Engineering, NLP, Regression & Classification, Evaluation & Benchmarking, Deep Learning, RAG, LLMs

Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, PyTorch, Hugging Face, TensorFlow, LangChain

Visualization & BI: Power BI, Tableau, Streamlit, Matplotlib

Databases: MySQL, FAISS, Pinecone

Languages: Python, SQL, C++

Tools: Jupyter Notebook, Google Colab, Git, Docker

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana